

→ can be parsed into two ways
unambiguous version:

$$\begin{aligned} E &\rightarrow E + T \mid T \\ T &\rightarrow T * F \mid F \\ F &\rightarrow id \end{aligned}$$

Ques)

Define chomsky normal form of grammar

⇒ A CFG is in chomsky normal form if each production is of the form:

- $A \rightarrow BC$ (two non-terminals)
- $A \rightarrow a$ (single terminal)

• optional $S \rightarrow \epsilon$ (only if ϵ is in language)

→ chomsky normal form is a specific way of writing a context free grammar where production rules follow a very restricted pattern.

Ques)

Show what languages are generated by the given context free grammars in each case

i). $S \rightarrow aSb \mid bSa \mid$

ii). $S \rightarrow SS \mid bS \mid a$